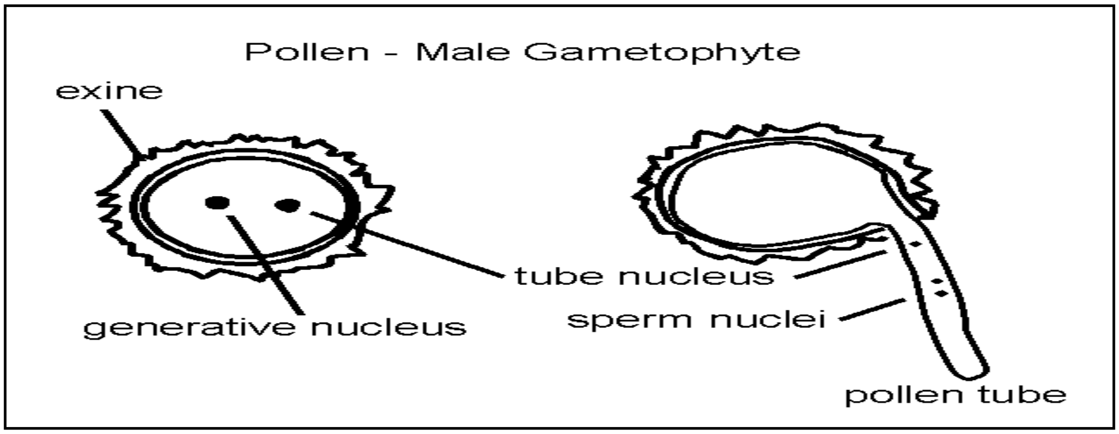
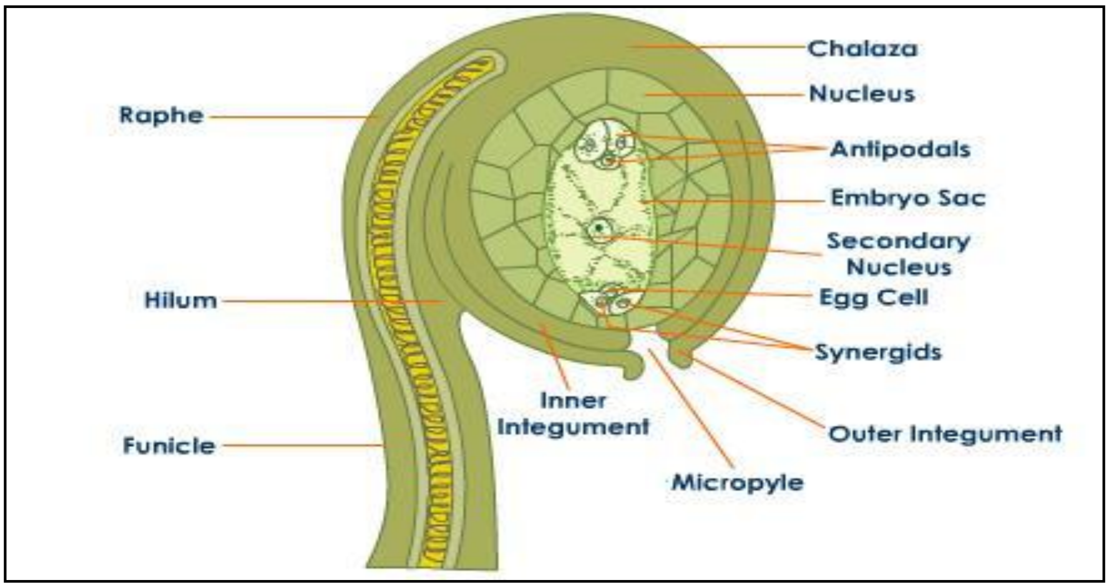
L No 8.
Reproduction in Plants

1. **Sketch & label typical angiospermic pollen grain. (2m)**



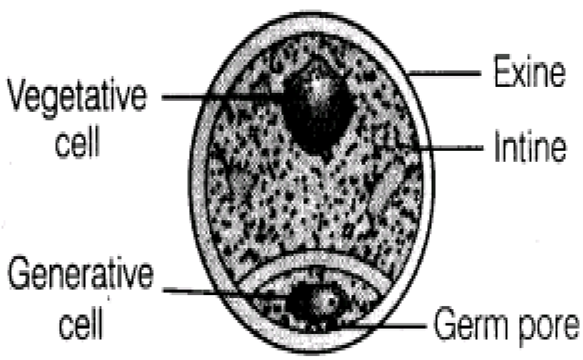
2. **With the help of neat labeled diagram describe the structure of anatropous ovule, Add a note on the functions of its various parts. (7m)**



1. The ovule which has a bent axis and downwardly directed micropyle is called anatropous ovule.
2. Mature anatropous ovules consist of 2 parts. i.e. stalk and the body.
3. The stalk of the ovule is called the funicle or funiculus.
   Function: the funicle attaches ovule to the placenta
4. The funicle bears a swollen part at its tip called body of the ovule.
5. The point at which funicle is attached to body of ovule is called hilum.
6. The body of ovule consists of 5 parts: nucellus, chalaza, integument, micropyle and embryo sac.
7. **Nucellus**: It represents the megasporangium proper of an ovule. It is made up of parenchymatous cells.
   Function: It helps in development of embryo sac.
8. **Chalaza**: The basal part of the nucellus is called chalaza. The integuments of ovule develop from chalaza.
9. **Integuments**: they are protective coverings of nucellus. They are not present at apical position.
   Function: The y give protection after fertilization they act as seed coat.
10. **Micropyle**: Micropyle is a small tip below at nucellus. **Function:**It provides entry for pollen tube.It also allow entry of water during fertilization.
11. **Embryo sac**: In mature ovule the nucellus shows an oval shaped structure towards its micropylar end called embryo sac or female gametophyte. It has 7 cells & 8 nuclei.

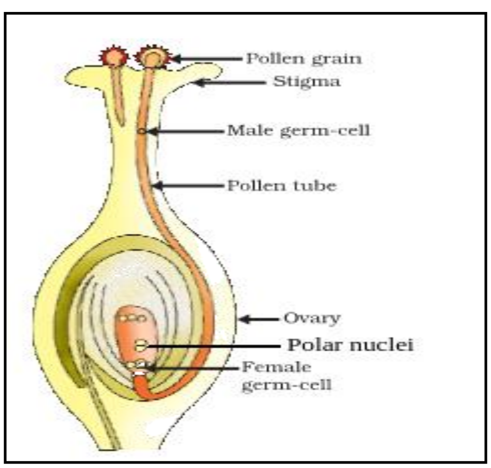
12. **Egg apparatus**: there is 3 celled egg apparatus lying at hemicropylar end consisting of median egg cell & 2 lateral cells called synergids.**Function:**After fertilization it gives rise to diploid zygote which develop into embryo.Synergids helps in fertilization.
13. **Antipodal cells**: At the opposite end of the egg apparatus towards the challazal end lies antipodal cells.**Function:They are accessory cells which degenerate after fertilization.**
14. **Secondary Nucleus**: In the middle of embryo sac lies a large central cell having 2 haploid polar nuclei which at a later stage fuse with each other to fom a diploid secondary nucleus.**Function:**It forms primary endosperm after double fertilization.
15. **Embryo sac**: It is generally monosporic, endosporic 7 celled, 8 nucleated structure. It is also called as polygonum type of embryo sac.

3. Describe the structure of angiospermic pollen grain. (3m)



1. A typical angiospermic pollen grain is unicellular, uninucleate, spherical or haploid structure.
2. It is also called microspore
3. It is covered by 2 layers outer exine and inner intine.
4. Exine is made up of sporopollenin and it protects pollen grain from biological & physical decomposition.
5. Exine is spiny in insect pollinated flowers and smooth in wind pollinated flowers.
6. Exines are discontinuous and interrupted by small pores called germ pores
7. Intine is composed of cellulose and pectin encloses the protoplasm with a single haploid nucleus.

4. Describe the process of double fertilization. Explain its importance. (7m)

Ans: 

<u>Double fertilization</u>

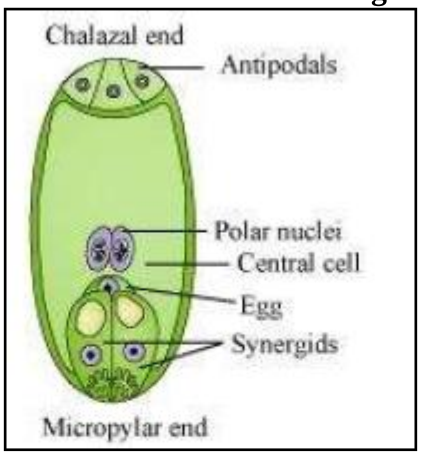
1. Out of 2 male gametes produced by the male gametophyte in angiosperms, one unite with the fgemale gamete and the other with secondary nucleus.

2. Since both the male gametes take part in fertilization and fertilization occurs twice , it is called double fertilization.
3. During double fertilization the pollen tube on reaching the ovule enters the embryo sac through micropyle and bursts.
4. Owing to this the 2 male gametes contained in pollen tube are set free.
5. Out of 2 male gametes one unites with the egg or female gamete and the other unite with the secondary nucleus of the embryo sac forming a triploid or triple fusion nucleus, called the primary endosperm nucleus.
6. The process involving the fusion of haploid nucleus of male gamete with the egg nucleus resulting in formation of diploid zygote is called syngamy.
7. The process involving the fusion of hgaploid nucleus of male gamete with diploid secondary nucleus to form a triploid primary endosperm nucleus is called triple fusion.
8. The reproductive process in which nonmotile male nuclei are carried to the egg cell through a pollen tube is called siphonogamy.
9. After fertilization certain changes take place in the ovule leading to the development of a seed.

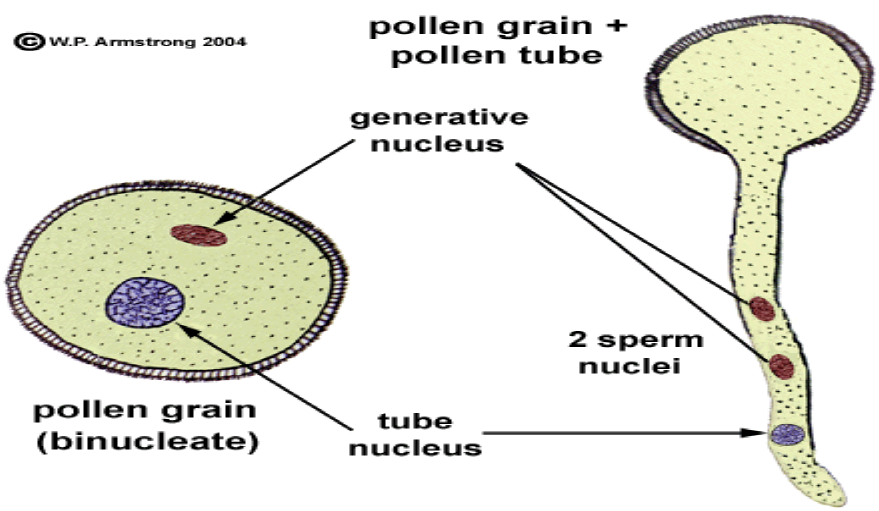
<u>Significance of double Fertilization:</u>

1. Double fertilization leads to the formation of triploid endosperm which ensures better nourishment of the developing embryo.
2. Double fertilization leads to the formation of an embryo which gives rise to a new plant.
3. Double fertilization is essential for formation of viable seeds.
4. Double fertilization restores diploid condition by the fusion of 2 haploid gametes.
5. Double fertilization brings about the recombinations of characters.
6. Double fertilization stimulates the ovule as a result of which the ovule is converted into the seed and the ovary is converted into the fruit.

5. Sketch and label female gametophyte of angiosperm. (3m)



6. Sketch and label male gametophyte of angiosperm. (3m)



7. Describe any 2 natural methods of vegetative propagation. (4m)

Ans: **Vegetative Propagation** : The reproduction which occurs with the help of vegetative oprgans like root, stem, leaves or buds is called vegetative propagation.

Natural methods of vegetative propagation

A]By tuberous roots: In some plants roots are swollen due to storage of food material. These tuberous roots bear adventitious buds on their surface which under favourable conditions sprout to produce leafy shoots and slips.It is seen in sweet potato, asperagus, dahlia.

B] By stem tubers: It is underground modification of stem. It bears number of notches on its surface called eyes. The eye is actually a node bearing one or more small axillartybudsand reduced scaly leaves.One of the axillary bud sprout and give rise to a new plant. Eg: potato

C] By runner: the runner is subaerial slender branch which creeps horizontallyon the ground and become rooted at the nodes. Shoots which are produced from upper isdes of the nodes get detached from the parent plant and grow as a independent plant. Example: cynodon, Fragaria, Oxalis.

D] By leaf buds: In plants like bryophyllum, Kalanchoe and Begonia vegetative propagation occurs with the help of leaf buds. In bryophillum leaf bears buds in its notched margins. These buds sprout on the leaf to form leafy shoots and adventitious roots which on separation fall on the wet soil and develop into a new plant.

8. Draw a neat well labeled diagram of Anatropous ovule. (3m)

Ans: refer Q No 2.

9. Give significance of double fertilization. (3m)

Ans: Significance of double Fertilization:

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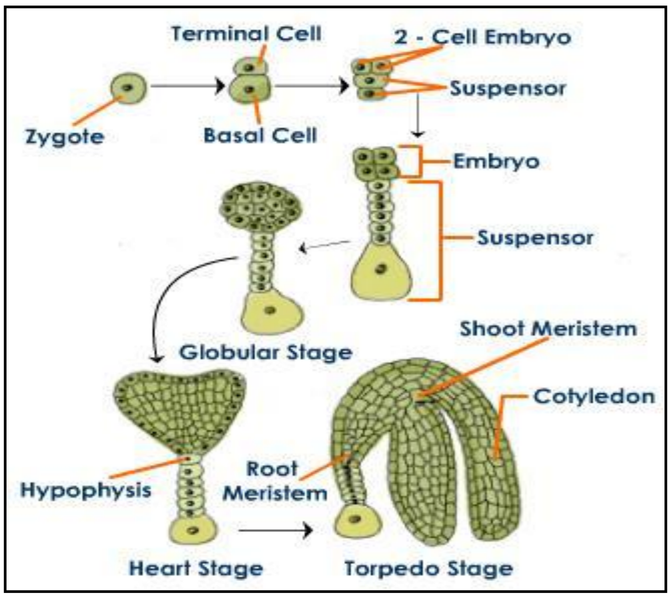
10. Explain the development of dicot embryo in angiosperms.

Ans:



1. The development of embryo from a zygote is called embryogenesis.

2. The fusion of male gamete and an egg cell during fertilization results in formation of diploid zygote.
3. The zygote develops a wall around it and is converted into oospore.
4. The oospore undergoes a transverse division to form a large basal cell towards the micropyle and a small apical cell towards the interior of the embryo sac. This 2 celled structure is called embryosac.
5. The basal cell undergoes repeated transverse divisons to form multicellular structure called suspensor.
6. The suspensor pushes the embryo towards the endosperm to draw its nutrition.
7. The apical cell of the proembryo undergoes a transverse division followed by 2 vertical divisions at right angle to form an octant stage.
8. The octant forms 8 smaller cells and 8 larger cells.
9. The cells of outer layer forms epidermis.
10. The cells of the inner layer gives rise to an embryonal mass which on differentiation form procambium and ground meristem.

11. The procambium and ground meristem undergo differentiation and form an embryonal axis consisting of plumule, a radical and 2 cotyledons.

11. Give functions of tapetum& Suspensor.

Ans: <u>Tapetum:</u>Tapetum is important for the nutrition and development of pollen grains, as well as a source of precursors for the pollen coat.

<u>Suspensor:</u>The suspensor helps in pushing the embryo into the endosperm. The **suspensor functions** early in embryogenesis to provide physical support, nutrition, and growth regulators for proper development of embryo.

12. Give floral adaptations of anemophily, ornithophily, chiropterophily&entamophily. (3m)

Ans: <u>Anemophily:</u>

1. The transfer of pollen grain through wind is called wind pollination.
2. Plants that are pollinated by wind are called as anemophilous plants.
3. Animophilous plants bear small & inconspicuous flowers without any bright colours fragrance and nectar.
4. Flowers are produced in large numbers
5. Stamens are long with versatile anthers.
6. Stigma is feathery to receive pollen grains coming along with the wind.
7. Example: grass, maize, Jowar& sugarcane.

<u>Ornithophily</u>:

1. The transfer of pollen grain through birds is called ornithophily.
2. Plants that are pollinated by wind are called as ornithophilous plants.
3. Ornithophilous plants bear large and fleshy flowers.
4. Flowers are brightly coloured to attract birds for pollination
5. Ornithophilous flowers lack fragrance as birds have poor sense of smell.
6. Pollen grains are sticky.
7. Example: Callistemon, Bignonia, Bombax, Butea.

<u>Chiroterophily;</u>

1. The transfer of pollen grain through bats is called ornithophily.
2. Plants that are pollinated by bats are called as chiroterophilous plants
3. Bat-pollinated flowers tend to be large and show, white or light coloured, open at night and have strong musty odours.
4. They are often large and bell-shaped or a ball of stamens.
5. Flowers are typically borne away from the trunk or other obstructions
6. Example:Adansonia, kigellia.

<u>Entamophily:</u>

1. The transfer of pollen grain through insects is called **entamophily**.
2. Plants that are pollinated by insects are called as **entamo**philous plants.
3. Flowers are large and attractive.
4. When small, flowers are clustered in a n inflorescence.
5. Flowers are brightly coloured with pleasant smell
6. Flower produce nectar which is food for insects.
7. Poleln grains are spiny and sticky for easy adherence to stigma.
8. Example: Hibiscus, rose, mogra.

[each pollination can be asked for 3/2 marks example is necessary, can be asked as diference between also]

<u>Hydrophily</u>

Hydro means water and phily means affinity. Hydrophily are water pollinated plants. One example is Vallisneria. The characteristics of water-pollinated plants are:
- Pollen grains are produced in large numbers.
-The weight of the pollen grains is equal to that of water. That is why they float on the surface of water.
-The pollen grains continue to float on the surface of water till they meet the female flowers. The same procedure is followed in the water pollinated flower called Vallisneria.

13. Distinguish between self & Cross pollination.

Ans:

Self pollination	Cross pollination
1. Autogamy	1.Allogamy
2. No agent required	2. Agent is required
3. Male & female parts mature at the same time	3. Male & female parts mature at different times
4. Can occur even if the flower is closed	4. Can occur only when flower is open
5. Offspring can become Weak	5. Offspring healthier
6. New variety not possible	6. New variety possible
7. Preserves parental Characters	7. Does not preserve parental characters

14. Write short note on advantages & disadvantages in cross pollination.

15. Ans: __Advantages of cross pollination:__

- **1. The offspring are healthier, disease resistant and better adapted to the climatic conditions.**
- **2. The seeds are produced in larger number and are more viable.**
- **3. The seeds develop and germinate properly and grow into better plants.**

4. Improved varieties can be developed through genetic recombination.

5. Through cross pollination desirable characters can be introduced and undesirable characters can be eliminated.

__**Disadvantages of cross pollination:**__

- **1. It is not always certain as a pollinating agent is always required, and it may or may not be available at the suitable time.**
- **2. Pollen grains have to be produced in abundance to ensure chances of pollination. This results in lot of wastage of pollen.**
- **3. It is uneconomical for plants as they have to produce flowers that are large, perfumed and with nectar to attract insects.**
- **4. Plants are unable to maintain genetic purity in their offspring.**
- **5. Owing to genetic recombination, desirable chacters may be eliminated and undesirable charcters may be introduced in the offsprings.**

16. Define geitonogamy&xenogamy. Give advantages of self-pollination. Explain how dichogamy favors cross pollination

Ans__: Geitonogamy__:- transfer of pollen grains from anthers to stigma of another flower of the same plant.

__**Xenogamy**__:-Transfer of pollen grains from anther to stigma of a different flower of different plants

__**Advantages of self-pollination**__

1. Self pollination is a sure method of pollination.
2. Flowers do not depend on external agents like wind, insects, etc for pollination

3. It is more economical because they do not have to develop large, attractive flowers, fragrance nectar & honey etc.
4. There is least wastage of pollen grains during self pollination.
5. Genetic stability is maintained in the progeny.
6. Pure line plants can be preserved through self pollination.
7. Improved varieties of plants can be multiplied by preserving their desirable characters.

- **Dichogamy favors cross pollination**
- Maturation of stigma and anther at different times is called dichogamy. As they do not mature together they can't self pollinate. Hence dichogamyfavours cross pollination.
- It is of 2 types:
- A] Protandry : In plants like sunflower pollen grains are mature much before stigma, such plants are called protandry.
- B] Protogyny: : In plants like Michelia stigma get mature much before pollen grains, such plants are called protogyyny..

17. Formation of primary endosperm nucleus is called triple fusion. Give Reason.

Ans: 1. Triple fusion is the fusion of the male gamete with two polar nuclei inside the embryo sac of the angiosperm.
2. This process of fusion takes place inside the embryo sac.
3. One male gamete nucleus and two polar nuclei are involved in this process.
4. When pollen grains fall on the stigma, they germinate and give rise to the pollen tube that passes through the style and enters into the ovule.
5. After this, the pollen tube enters one of synergids and releases two male gametes there.
6. Out of the two male gametes, one gamete fuses with the nucleus of the egg cell and forms the zygote (syngamy).
7. The other male gamete fuses with the two polar nuclei present in the central cell to form a triploid primary endosperm nucleus.
8. Since this process involves the fusion of three haploid nuclei, it is known as triple fusion. It results in the formation of the endosperm.

18. Sketch & label T. S of angiospermic anther.

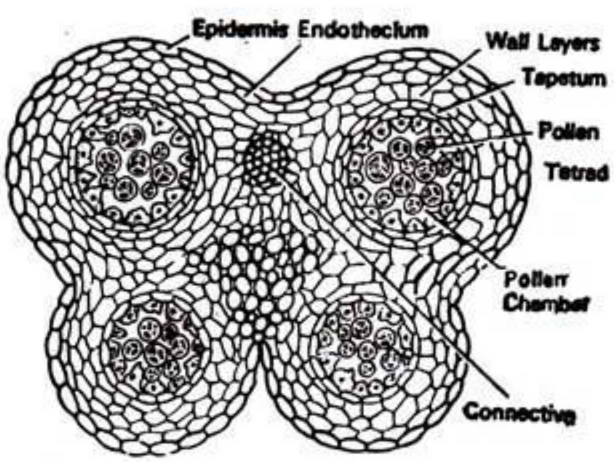


Fig. 182. T.S. anther with mature pollen sacs.

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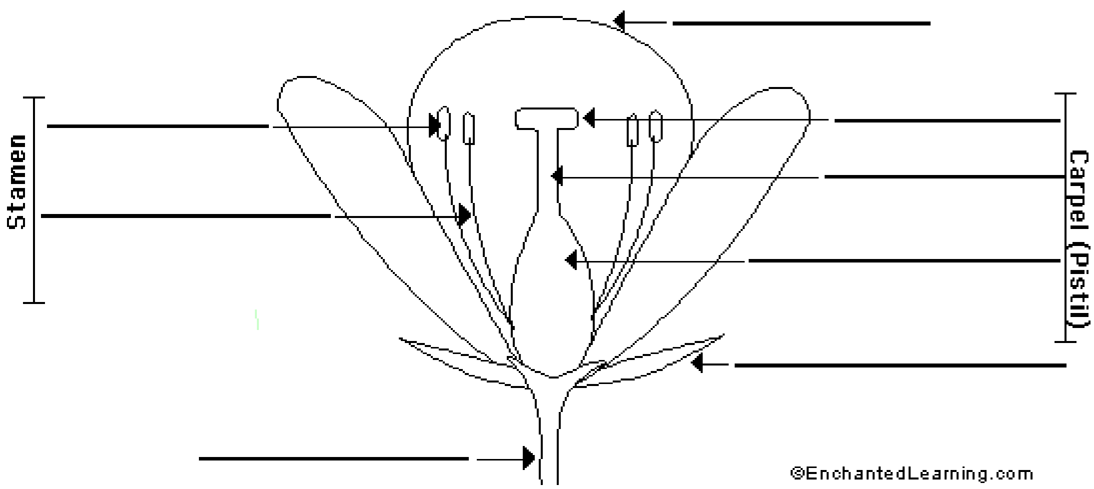
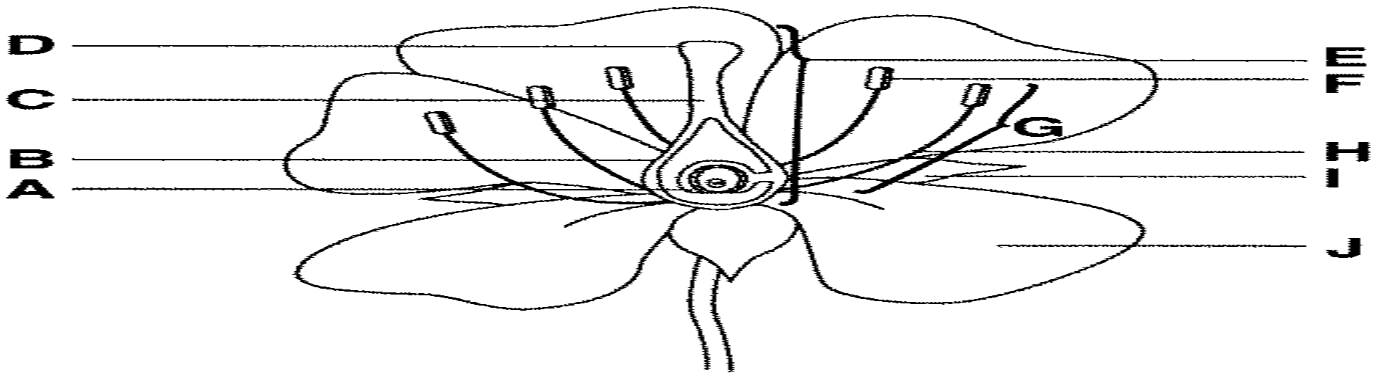
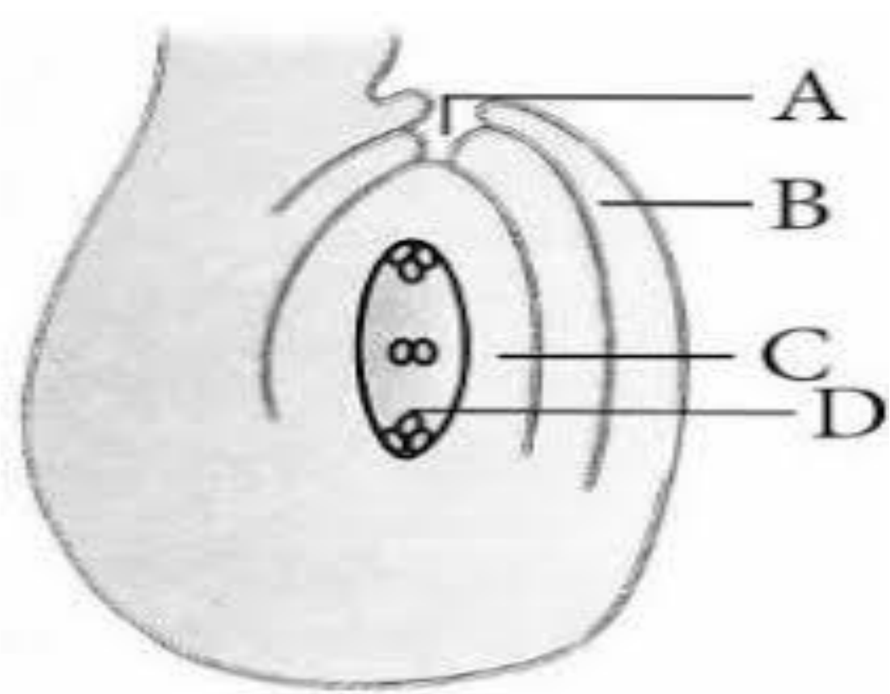
Assignment

1. In the given diagram label the following parts.

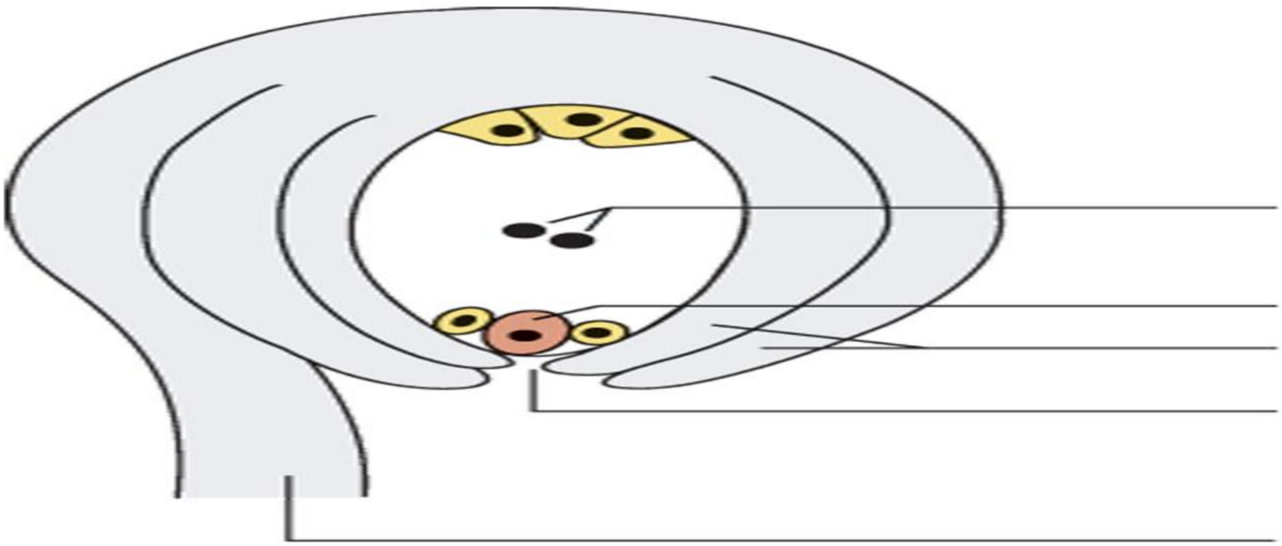




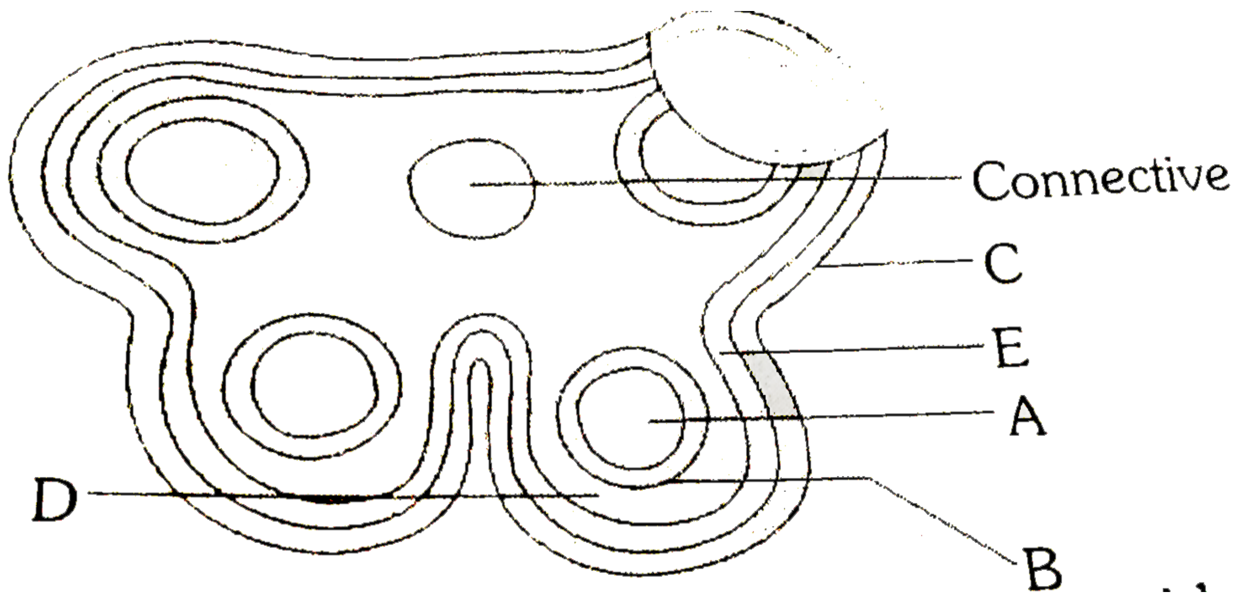
Eat Label the structure of anatropus ovule



2. Label the structure of embryo sac



3. <u>Label the structure of T S of anther</u>



4. Structure of Pollen grain